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FitWell: Fitness + Wellness Database

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Executive Summary

FitWell is an all-encompassing fitness/wellness database that is aimed toward anybody who wants to track their fitness journey or wellness goals. Fitwell is intended to be user-friendly and simple to use so that the entry barrier is low and users do not get overwhelmed when interacting with the functions of the database.

Vision and Objective

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The objective of FitWell database system is designed so that users can track multiple aspects of their fitness and wellness journey, including goals, workout plans, specific exercises, workout days, nutrition, supplements, and logs. An all in one database where users can keep every aspect tracked and recorded with ease.

Mission Statement

The purpose of FitWell is to make the hassle of tracking workouts, finding workouts and exercises that work, staying on top of your diet a better time. As somebody who is into weightlifting, I still struggle with all these aspects. I want to make all these processes simple to interact with to make peoples lives a little easier when doing this stuff.

Service

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The services that FitWell offers are a workout selector/builder where users can pick or design their own workouts, from a variety of exercises to choose from. A diet planner and tracker where users can track their eating throughout the day and find diets that work for them. A log where users can log their workouts to see the pounds go up on the bar.

Use Case

A user would load into the application and see the sign up page where they would be prompted into signing up with an account. After the signup is complete they are offered a guided tutorial that asks questions about what workouts they want to do and what diet they want to do. This tutorial is skippable and the app ensures that there is no pressure to rush into everything at once. Once completed then the user has their home page that acts like a dashboard of all the things that they are currently logging. There are individual tabs for workouts, nutrition and logs.

Business Requirements

The FitWell system must satisfy several functional and non-functional business requirements to ensure usability and effectiveness.

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1. Users must be able to create and manage personal accounts.
2. The system must allow users to create, edit, and select workout plans.
3. Only one workout plan can be active for a user at a time.
4. Workout plans must be completed before being marked as active.
5. The system must store detailed information about exercises, including categories and difficulty levels.
6. Users must be able to log workouts and track exercise performance over time.
7. Users must be able to track nutrition, including meals and food items.
8. The system must support supplement tracking and logging.
9. Users must be able to record and monitor personal records for exercises.
10. The system must provide a reporting feature for users to submit issues or feedback.
11. Data integrity must be maintained through the use of primary and foreign key constraints.
12. The system must be scalable to allow additional features in future releases.

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Entity-Relational Model Diagram

The ER diagram for the FitWell database illustrates the relationships between all entities within the system. The Users table serves as the central entity, connecting to multiple components such as goals, workout plans, nutrition logs, supplement tracking, and progress entries.

One-to-many relationships exist between Users and entities such as Goals, WorkoutLogs, NutritionLogs, ProgressEntries, and Reports. Many-to-many relationships are resolved using bridge tables, including WorkoutDayExercises, WorkoutLogExercises, MealFoods, UserWorkoutPlans, and UserSupplements.

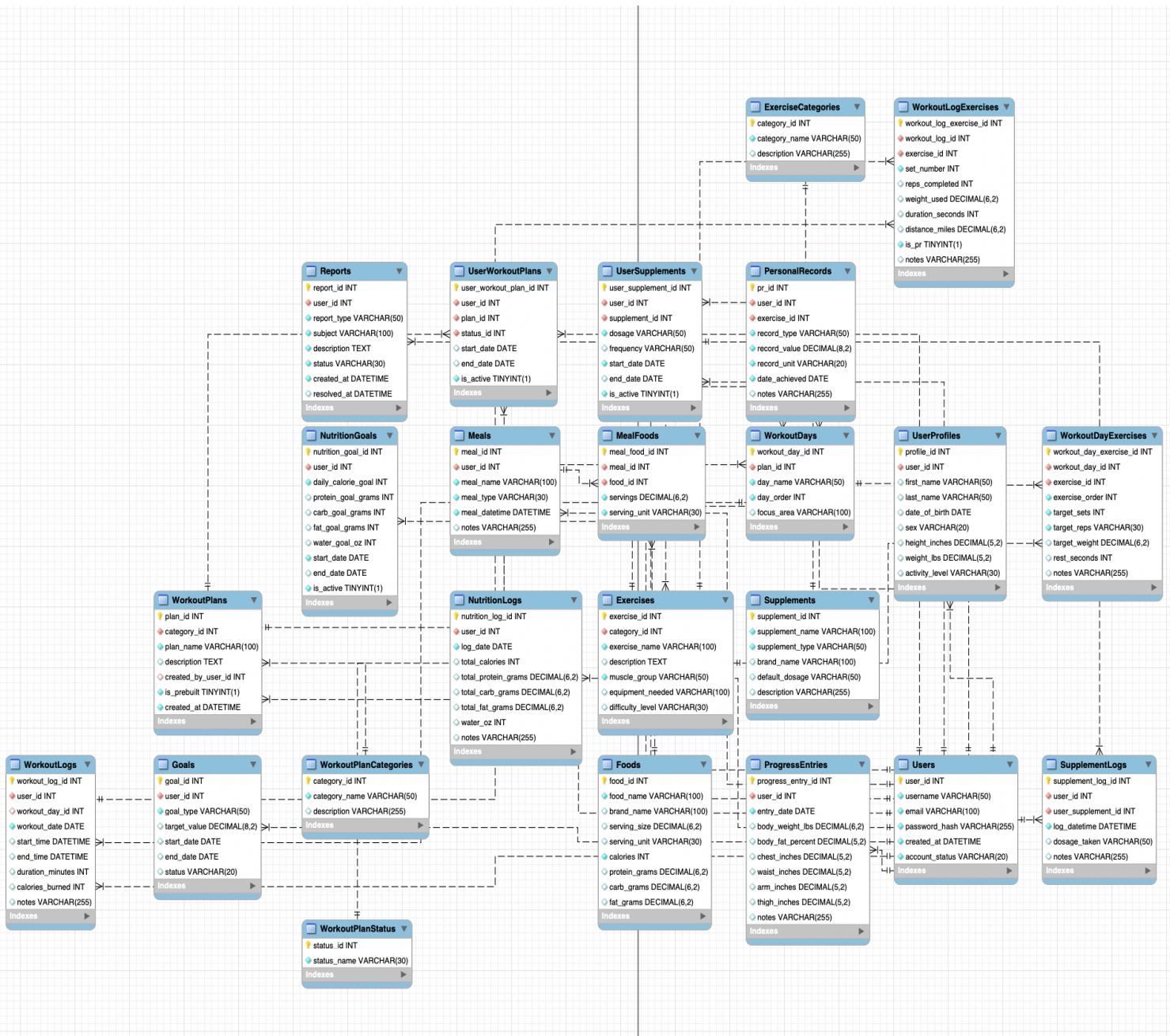
WorkoutPlans are categorized using WorkoutPlanCategories and assigned to users through UserWorkoutPlans. Each workout plan consists of multiple WorkoutDays, which are further linked to Exercises through WorkoutDayExercises.

Nutrition tracking is implemented through Meals and Foods, connected via the MealFoods table, while supplement usage is managed through UserSupplements and SupplementLogs.

All relationships are enforced using foreign key constraints to maintain referential integrity and ensure consistency across the database.

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ER Diagram



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Development Approach

The FitWell database was developed using a structured relational design approach. The process began with identifying the key features required for a fitness and wellness tracking system. Core entities such as users, workouts, exercises, nutrition, and supplements were defined based on these requirements.

An initial ER diagram was created to visualize relationships between entities. Special attention was given to normalization and eliminating redundancy. Many-to-many relationships were resolved using bridge tables to ensure efficient data storage and scalability.

The design was implemented using MySQL Workbench, where tables were created with appropriate data types, primary keys, and foreign key constraints. The database was then populated with sample data to test functionality and validate relationships between entities.

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Current Release

The current version of FitWell includes a fully functional database schema that supports all core features. Users can create accounts, manage profiles, define fitness goals, and interact with workout plans.

The system allows users to log workouts and track exercise performance over time. Nutrition tracking is supported through meals and food entries, while supplement usage is tracked through dedicated tables.

Additionally, users can record personal records and monitor progress using progress entries. All data is structured and stored using relational tables, ensuring consistency and reliability through enforced relationships.

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Future Releases

Future improvements to FitWell will focus on expanding functionality and improving usability. One major enhancement would be the development of a user interface, such as a web or mobile application, that connects directly to the database.

Additional features may include data visualization tools such as progress charts, personalized workout and nutrition recommendations, and integration with external fitness or nutrition APIs.

Further development could include social features, allowing users to share progress, as well as advanced analytics to provide deeper insights into user performance and habits.

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Conclusion

The FitWell database demonstrates the successful design and implementation of a comprehensive fitness and wellness tracking system. By integrating multiple aspects of a user's fitness journey into a single database, the system provides a structured and efficient solution for managing data.

Through the use of relational database principles, including normalization, primary and foreign keys, and bridge tables, the system ensures data integrity and scalability. This project highlights the importance of well-structured database design and provides a strong foundation for future expansion into a full application.

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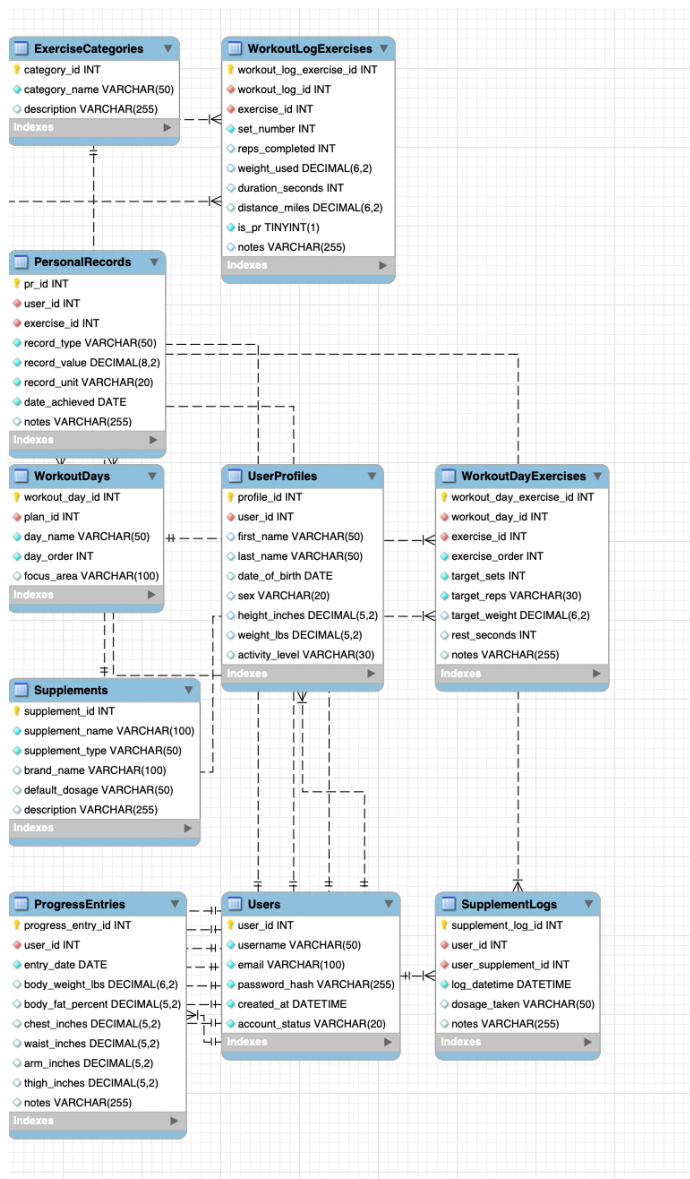
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Appendix:

- ER Diagram (zoomed images)

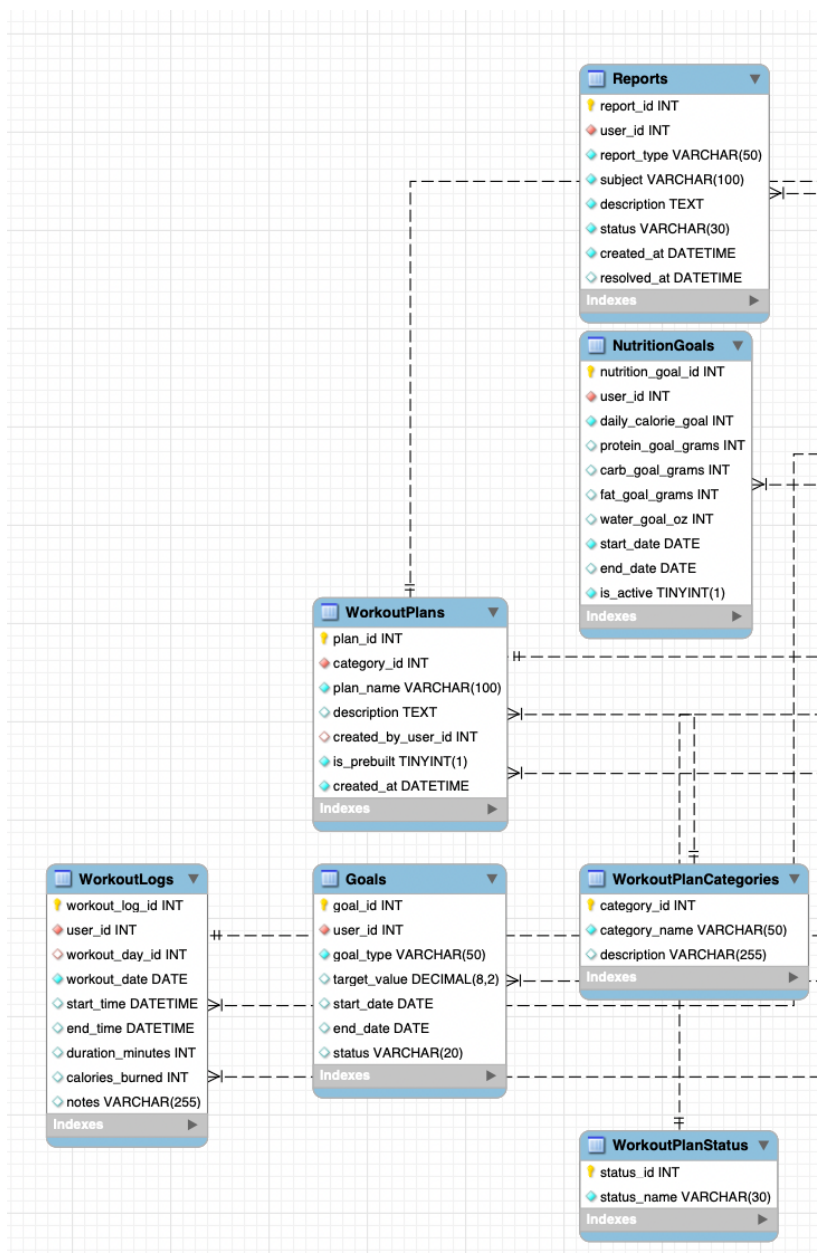


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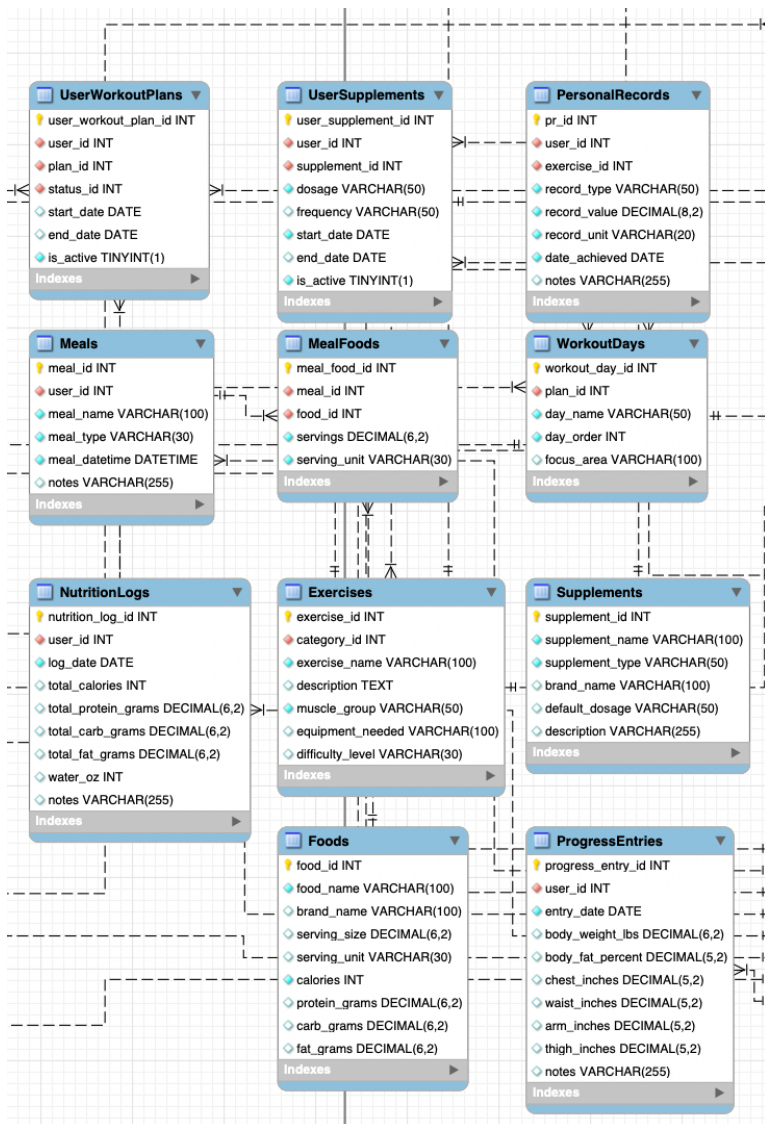


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- Sample table outputs (SELECT query screenshots)

Users Table

[illegible]

WorkoutPlans Table

[illegible]

Exercises Table

[illegible]

- SQL scripts (CREATE TABLE and INSERT)
- fitwell_create_tables.sql
- fitwell_insert_data.sql

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- Database export file (fitwell_dump.sql)

- fitwell_dump.sql